

Education

IRIT Laboratory, University of Paul Sabatier

Toulouse, France

PH.D. IN COMPUTER SCIENCE

2019 - 2022

- Thesis Topic: Studying Relevant Signals for Document Retrieval using Transformer Models
- Highlights: Enhancing, fine-tuning, and evaluating encoder models for ad hoc retrieval based on cross-encoder and dual-encoder architectures, using single-vector and multi-vector representations, using direct supervision or distillation from teacher models.
- Advisors: Jose G. Moreno, Mohand Boughanem
- Area of study: Deep Learning, Information Retrieval, and Natural Language Processing

Higher National School of Computer Science (ESI)

Algiers, Algeria

MASTER'S DEGREE IN COMPUTER SCIENCE AND ENGINEERING

2018 - 2019

- Thesis Topic: Real Time Tweet Summarization
- Advisors: Lynda Said Lhadj, Mohand Boughanem
- Area of study: Deep Learning, Information Retrieval, and Natural Language Processing

Higher National School of Computer Science (ESI)

Algiers, Algeria

ENGINEERING DEGREE IN COMPUTER SCIENCE AND ENGINEERING (VALEDICTORIAN)

2014 - 2019

- Majored in Information Systems & Software

Experience

Mindflow

Paris, France

NLP RESEARCHER

oct. 2023 - present

- Designing no-code solutions powered by Generative AI agents to streamline task orchestration in cybersecurity operations.
- Designed and deployed multi-agent systems leveraging Large Language Models (LLMs) and Retrieval-Augmented Generation (RAG), enabling seamless natural language interactions with over 3,000 API providers. [Patent Pending]
- Evaluated different retrieval approaches for RAG experimenting with sparse, dense, and hybrid models to improve retrieval accuracy and reduce latency.
- Studied various multi-agent workflows, including collaboration through shared memory and hierarchical agent interactions to enhance scalability and reduce cost.
- Designed an evaluation framework for assessing the planning and tool-use capabilities of LLMs, ensuring high-quality performance and decision-making in complex scenarios.

LIG, University of Grenoble Alps

Grenoble, France

POSTDOC RESEARCH FELLOWSHIP

dec. 2022 - sept. 2023

- Enhancing the systematic compositional generalization capabilities of sequence-to-sequence (seq2seq) models by integrating structural (syntactic) information into the decoding process through hyperbolic representations of dependency trees.
- Designed, trained, and evaluated a hybrid Euclidean-Hyperbolic transformer-based seq2seq model from scratch.

Research Internships

Max Planck Institute for Informatics (MPI)

Saarbrücken, Germany

A STUDY OF TERM-TOPIC EMBEDDINGS

Jun. 2021 - Feb. 2022

- Advisor: Andrew Yates
- Studied advancements with the ColBERT architecture, which relies on token-level representations with late interactions for document ranking.
- Proposed a novel approach for distilling ColBERT's contextualized token embeddings into a more explicit and principled process by aggregating a finite set of frozen, pre-trained term-topic embeddings, with each term-topic capturing a contextual topic of a token.

Institut de Recherche en Informatique de Toulouse (IRIT)

Toulouse, France

RESEARCH ON MICROBLOG RETRIEVAL AND SUMMARIZATION

Dec. 2018 - Jul. 2019

- Advisor: Mohand Boughanem
- Development of a tweet summarization approach based on Deep Learning Models. Manipulation of various state-of-the-art Models for both tweet representation and relevance estimation of tweets with respect to users' interests.

Publications

- Lila Boualili
Deep learning for information retrieval: studying relevant signals for ad hoc search based on transformer models
Thesis, Université Paul Sabatier-Toulouse III
- Lila Boualili and Andrew Yates
A Study of Term-Topic Embeddings for Ranking
Advances in Information Retrieval: 45th European Conference on Information Retrieval, ECIR 2023, Dublin, Ireland, April 2–6, 2023, Proceedings, Part II
- Lila Boualili, Jose G. Moreno, and Mohand Boughanem
Highlighting Exact Matching via Marking Strategies for Ad hoc Document Ranking with Pretrained Contextualized Language Models
Information Retrieval Journal 25 (4), 414-460
- Lila Boualili, Jose G. Moreno, and Mohand Boughanem
MarkedBERT: Integrating Traditional IR Cues in Pre-trained Language Models for Passage Retrieval
Proceedings of the 43rd International ACM SIGIR Conference on Research and Development in Information Retrieval (SIGIR '20), X'ian, China, July 25-30, 2020.
- Lila Boualili, Lynda Said Lhadj, and Mohand Boughanem
Retrospective Tweet Summarization: Investigating Neural Approaches for Tweet Retrieval
Proceedings of the Joint Conference of the Information Retrieval Communities in Europe (CIRCLE 2020), Samatan, Gers, France, July 6-9, 2020.

Projects

DEEPRESEARCHPY

- A Python package for automated deep query investigation, adapted from JinaAI's node-deepsearch, it iteratively searches, reads, and reasons across the web until finding a satisfactory answer or reaching a token budget limit.

SCIWATCH

- A Python package that delivers scheduled newsletters with relevant scientific papers, using boolean retrieval for query matching.

HYBRIDSEQ2SEQ

- Transformer-based Euclidean-hyperbolic hybrid seq2seq model for COGS semantic parsing, leveraging hyperbolic embeddings to capture hierarchical structures and enhance compositional generalization.

Teaching Activities

University of Paul Sabatier

Toulouse, France

IN FRENCH.

2019 - 2021

- Data Structures and Fundamental Algorithms (Semester I)
- Database Programming and administration (Semester II)

Skills

Programming	Python, TypeScript, rust
Libraries	Pytorch, TensorFlow, Sickit-learn, Transformers, Sentence-Transformers, Langchain, Fairseq
Operating Systems	Microsoft Windows, Linux and other UNIX variants
Agile Methodologies	Scrum and Kanban
Languages	French (native), English (professional), Arabic (professional), Kabyle (native)

Hobbies

- Avid reader with a strong interest in Lovecraftian literature.
- I enjoy drawing and I am exploring photography as a new creative outlet.